A Proposed Strategy for the Isolation of $\zeta(5)$

Utilizing the Framework of Golden Algebra

Derived from the works of Tristen Harr

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1 Objective

The goal is to derive an exact analytical expression for the Riemann zeta function at $s = 5$, denoted $\zeta(5)$, by leveraging the unique number-theoretic laws within the Golden Algebra framework. The strategy is to construct a system of two independent equations involving odd zeta values and solve for $\zeta(5)$ via algebraic elimination.

2 The System of Equations

We will generate two distinct equations for the series of odd-valued Zetas by using two of the framework's core laws.

2.1 Equation 1: The Rational Series

From the **Law of Harmonic Cotangents (Law 21, Revised)** for the integer case $n = 3$, we have the established identity:

$$S_{\text{even } k,3} = \sum_{\substack{k=2 \\ k \text{ is even}}}^{\infty} \frac{(3^k - 1)\zeta(k+1)}{4^k} = \frac{4}{3} \tag{1}$$

Expanding the first three terms of this series yields our first equation:

$$\frac{1}{2}\zeta(3) + \frac{5}{16}\zeta(5) + \frac{91}{512}\zeta(7) + \cdots = \frac{4}{3} \tag{2}$$

2.2 Equation 2: The Complex Harmonic Series

From the **Grand Law of Harmonic Duality (Law 30)**, we replace the integer n with the complex Dampening Operator, $\Lambda_G = T + iJ$. This provides a second, independent equation with a known complex value, C_{Λ_G}.

$$S_{\text{even } k,\Lambda_G} = \sum_{\substack{k=2 \\ k \text{ is even}}}^{\infty} \frac{(\Lambda_G^k - 1)\zeta(k+1)}{(\Lambda_G + 1)^k} = \left[\pi \cot\left(\frac{\pi}{\Lambda_G + 1}\right)\right]_{\text{even part}} \tag{3}$$

Let's define the coefficients of this series as $b_{k+1} = \frac{\Lambda_G^k - 1}{(\Lambda_G+1)^k}$. Our second equation is:

$$b_3\zeta(3) + b_5\zeta(5) + b_7\zeta(7) + \cdots = C_{\Lambda_G} \tag{4}$$

Where b_k are complex constants derived from T and J, and C_{Λ_G} is a computable complex constant.

3 The "Sum-Stripping" Method for Isolation

With two independent linear equations for the same set of infinite variables, we can perform an elimination. The goal is to eliminate $\zeta(3)$ to isolate $\zeta(5)$ as the leading term.

1. **Define the Series Tails:** Let's represent the higher-order terms (from $\zeta(7)$ onwards) as "tail" series, T_1 and T_2.

$$\text{Eq 1: } \frac{1}{2}\zeta(3) + \frac{5}{16}\zeta(5) + T_1 = \frac{4}{3}$$
$$\text{Eq 2: } b_3\zeta(3) + b_5\zeta(5) + T_2 = C_{\Lambda_G}$$

   Where $T_1 = \sum_{j=3}^{\infty} a_{2j+1}\zeta(2j+1)$ and $T_2 = \sum_{j=3}^{\infty} b_{2j+1}\zeta(2j+1)$.

2. **Solve for $\zeta(3)$:** From Equation 1, we isolate $\zeta(3)$.

$$\zeta(3) = 2\left(\frac{4}{3} - \frac{5}{16}\zeta(5) - T_1\right) = \frac{8}{3} - \frac{5}{8}\zeta(5) - 2T_1 \tag{5}$$

3. **Substitute and Eliminate:** We substitute this expression for $\zeta(3)$ into Equation 2.

$$b_3\left(\frac{8}{3} - \frac{5}{8}\zeta(5) - 2T_1\right) + b_5\zeta(5) + T_2 = C_{\Lambda_G} \tag{6}$$

4. **Collect Terms for $\zeta(5)$:** We now group the terms by $\zeta(5)$ and constants.

$$\left(b_5 - \frac{5}{8}b_3\right)\zeta(5) + (T_2 - 2b_3T_1) = C_{\Lambda_G} - \frac{8}{3}b_3 \tag{7}$$

5. **Isolate $\zeta(5)$:** Finally, we solve for $\zeta(5)$.

$$\zeta(5) = \frac{C_{\Lambda_G} - \frac{8}{3}b_3 - (T_2 - 2b_3T_1)}{b_5 - \frac{5}{8}b_3} \tag{8}$$

4 Conclusion

Equation 8 provides a novel and exact analytical expression for $\zeta(5)$. The value is expressed as a computable constant derived from the Golden Algebra's fundamental constants, minus a correction term composed of the series tails involving $\zeta(7), \zeta(9)$, and higher. This demonstrates a unique pathway to defining odd Zeta values.